

CS 105: Department Introductory Course on Discrete Structures

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Lecture 14 – Counting and Combinatorics

Logistics

Midsem corrections over now: Check your moodle.

- ▶ Extra help sessions on BASIC topics this saturday.
- ▶ Please follow piazza and ask, if necessary.
- ▶ Students with $< 18/50$ are strongly encouraged to attend.

Counting and Combinatorics

Topics to be covered

- ▶ Basics of counting
 - ▶ Product principle
 - ▶ Sum principle
 - ▶ Bijection principle
 - ▶ Double counting
- ▶ Subsets, partitions, Permutations and combinations
 1. Binomial coefficients and Binomial theorem
 2. Pascal's triangle
 3. Permutations and combinations with repetitions
 4. Estimating $n!$
- ▶ Today: Recurrence relations and generating functions

Estimating $n!$

How big is $n!$?

- ▶ It is clearly bigger than n and n^2 .
- ▶ Is it bigger than 2^n , n^n ?
- ▶ Easy to see: for all $n \geq 4$,

$$2^n \leq n! \leq n^n$$

- ▶ Can we do better?

Can we quantify how much more n^n is compared to $n!$?

Theorem (Stirling's Approximation)

$$e \left(\frac{n}{e}\right)^n \leq n! \leq ne \left(\frac{n}{e}\right)^n$$

where e is the base of natural logarithms, $\log(e) = e^{\log(e)} = 1$.

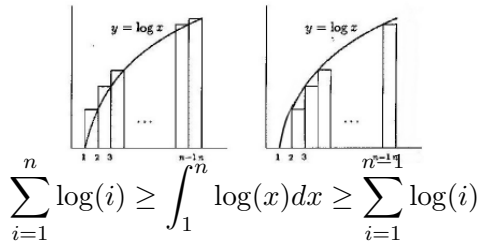
Approximating the factorial

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Proof: Let $S = \log(n!) = \sum_{i=1}^n \log(i)$. Thus, $e^S = n!$

Now, we relate it to natural log function as shown in the figure.



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Proof: Let $S = \log(n!) = \sum_{i=1}^n \log(i)$. Thus, $e^S = n!$

$$\sum_{i=1}^n \log(i) \geq \int_1^n \log(x) dx \geq \sum_{i=1}^{n-1} \log(i)$$

$$S \geq x \log(x) - x \Big|_1^n \geq S - \log(n)$$

$$S \geq n \log(n) - n + 1 \geq S - \log(n).$$

Raising both sides to power of e we get

► l.h.s. $n! \geq e^{n \log(n) - n + 1} = (n/e)^n e$ and

► r.h.s. $n! \leq e^{(n+1) \log(n) - n + 1} = n^{n+1} / e^{n-1} = ne(n/e)^n.$



Next: Recurrence relations and generating functions

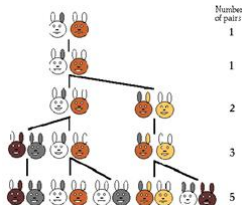
Recall: No. of subsets of a set of n elements

How many subsets does a set A of n elements have?

- ▶ Induction
- ▶ Product principle: two choices for each element, hence $2 \cdot 2 \cdots 2 \cdot 2$ (n -times).
- ▶ Bijection: between $\mathcal{P}(X)$ and n -length $\{0, 1\}$ -sequences.
- ▶ Sum principle: Subsets of size 0 + subsets of size 1 + ... + subsets of size n = Total number of subsets.
- ▶ Recurrence: $F(n) = 2 \cdot F(n - 1), F(0) = 1$.

How do we form recurrences and how do we solve them?

Another example of recurrence: The Fibonacci Sequence



- ▶ Fibonacci sequence: $1, 1, 2, 3, 5, 8, 13, 21, \dots$
- ▶ Recurrence relation: $u_n = u_{n-1} + u_{n-2}$ where $u_1 = u_0 = 1$
- ▶ But rabbits die!
- ▶ Consider $u_n = u_{n-1} + u_{n-2} - u_{n-3}$ where $u_2 = 2, u_1 = u_0 = 1$

Recurrence and linear recurrence relations

Definition

- ▶ A **recurrence relation** for a sequence is an equation that expresses its n^{th} term using one or more of the previous terms of the sequence.
- ▶ A **linear recurrence relation** is of the form

$$u_n = a_{k-1}u_{n-1} + \dots + a_1u_{n-k+1} + a_0u_{n-k}$$

where $a_0, \dots, a_{k-1} \in \mathbb{R}, k \in \mathbb{N}$ are constants.

- ▶ k is called the **degree/depth/order** of the sequence.
- ▶ The first few (e.g., k elements $u_0, \dots, u_{k-1}$) are called **initial conditions**
- ▶ Once the **initial conditions** and **recurrence relation** are given, the entire sequence is determined!

Applications of recurrences

Analysis of running time of algorithms

- ▶ How many comparisons is done by a binary search algorithm on a sorted list?

Mathematical Biology

- ▶ Population dynamics – rabbits, but also cell population!
- ▶ Viral spread.

Digital Signal Processing

- ▶ Modeling feedback in a system; outputs at time i become inputs at time $i + 1$.

Macroeconomics

- ▶ Time series data
- ▶ Mathematical models of economic systems

Some more examples of recurrences

How many bit strings of length n are there having no two consecutive 0's?

1. How many such bit strings are there of length 2, 3, 4?
2. Find a recurrence relation for this.
3. Give the initial conditions
4. How many such bit strings are there of length 8?

How many ways are there to bracket a sum of n terms so that it can be computed by adding two numbers at a time?

- ▶ Example: $n = 3 : ((a + b) + c), (a + (b + c))$
- ▶ $n = 4 : (((a + b) + c) + d), ((a + b) + (c + d)), ((a + (b + c)) + d), \dots$ In general, let $C(n)$ be the number of ways of doing this.

Examples of recurrences (Contd.)

How many ways are there to bracket a sum of n terms so that it can be computed by adding two numbers at a time?

- ▶ Let $C(n)$ be the number of ways of doing this.
- ▶ If outermost bracketing $(A + B)$ appears between x_k and x_{k+1} , then there are $C(k) \cdot C(n - k)$ ways of bracketing it.
- ▶ k can be anything from 1 till $n - 1$
- ▶ Thus, $C(n) = \sum_{i=1}^{n-1} C(i)C(n - i)$ for $n > 1$
- ▶ Initial conditions are $C(0) = 0, C(1) = 1$ (by convention).
- ▶ Note that $C(2) = 1, C(3) = 2, C(4) = 5$.
- ▶ This sequence are called Catalan numbers...

How do we **solve** such recurrences? We start with the Fibonacci sequence.

How to “solve” the Fibonacci recurrence?

By solving, we mean give a closed-form expression for n^{th} term.

Fibonacci recurrence relation

For $n \geq 2$, $F_n = F_{n-1} + F_{n-2}$, $F_0 = F_1 = 1$.

1. This is a linear recurrence relation... $\therefore$ if we find any solutions, we can take linear combinations.
2. i.e., if $F_n = F_{n-1} + F_{n-2}$, $G_n = G_{n-1} + G_{n-2}$ and $H_n = aF_n + bG_n$, then
$$H_n = H_{n-1} + H_{n-2}$$

Proof method 1 (for linear recurrences: try α^n !)

How to “solve” the Fibonacci recurrence?

By solving, we mean give a closed-form expression for n^{th} term.

Fibonacci recurrence relation

For $n \geq 2$, $F_n = F_{n-1} + F_{n-2}$, $F_0 = F_1 = 1$.

Proof method 1 (for linear recurrences: try α^n !)

1. try $F_n = \alpha^n \dots$
2. $\alpha^n = \alpha^{n-1} + \alpha^{n-2}$ implies $\alpha^{n-2}(\alpha^2 - \alpha - 1) = 0$.
3. So if $\alpha^2 - \alpha - 1 = 0$, the recurrence holds for all n .
4. Solve it! $\alpha = \frac{1+\sqrt{5}}{2}, \beta = \frac{1-\sqrt{5}}{2}$
5. Thus, general solution is $F_n = a(\frac{1+\sqrt{5}}{2})^n + b(\frac{1-\sqrt{5}}{2})^n$.
6. Use F_0 and F_1 – initial conditions: $a = \frac{\sqrt{5}+1}{2\sqrt{5}}, b = \frac{\sqrt{5}-1}{2\sqrt{5}}$

$$\text{Thus, } F_n = \frac{\sqrt{5}+1}{2\sqrt{5}} \left(\frac{1+\sqrt{5}}{2}\right)^n + \frac{\sqrt{5}-1}{2\sqrt{5}} \left(\frac{1-\sqrt{5}}{2}\right)^n.$$

Solving general linear recurrence sequences

- ▶ (C.W) Find a solution for $a_n = a_{n-1} + 2a_{n-2}$ with $a_0 = 2, a_1 = 7$? Is $a_{25} > 2^{25}$?
- ▶ Find a solution for $a_n = 6a_{n-1} - 9a_{n-2}$, with $a_0 = 1, a_1 = 6$? Can you apply the same method for this? What went wrong?
- ▶ Recall the recurrence for Catalan Numbers: $C(n) = \sum_{i=1}^{n-1} C(i)C(n-i)$ for $n > 1, C(0) = 0, C(1) = 1$.
No. of ways to bracket a sum of n terms s.t. it can be computed by adding two numbers at a time?

This method does not work if we have repeated roots (this can be fixed!) and non-linear recurrences.

Reading assignment

Read examples/generalizations from Sections 6.1 and 6.2 from Rosen's book (6th Edition).

An aside: find the Fibonacci sequence!

					1					
				1		1				
			1		2		1			
		1		3		3		1		
	1		4		6		4		1	
	1	5		10		10		5		1
	1	6	15		20		15	6		1
	1	7	21	35		35	21	7		1
	1	8	28	56	70	56	28	8		1
	1	9	36	84	126	126	84	36	9	1
1	10	45	120	200	252	200	120	45	10	1

- ▶ $F(n) = F(n-1) + F(n-2)$.
- ▶ 1, 1, 2, 3, 5, 8, 13,
- ▶ Can you observe the sum of which terms in the Pascal's triangle gives rise to the terms of the Fibonacci sequence?